Magppie HR App — Handover

In-house HR application to replace Keka · phone-first · prepared 21 October 2026

Code: github.com/jiyasiwach/magppie-hr-app (temporary home) · Live: magppie-hr-app.vercel.app · Local: http://localhost:3040

1. Summary

An HR app for Magppie's office, showroom, factory and site staff. The **front end is complete** — 33 screens covering attendance, leave, approvals, the employee directory, documents, policies, a company wall, surveys, reports, an HR assistant and a confidential grievance module. It currently runs on **invented sample data**; the database design is written but not yet switched on.

New in this round: attendance now requires the employee's **exact location** and a **face check** before any clock-in, clock-out or punch is saved, and the app has a new **white and sky-blue** look.

Area	Status	Notes
Screens / front end	**Complete**	33 pages, phone and desktop layouts
Location + face check on attendance	**Built and tested**	New this round — see section 3
Data	**Sample data only**	40 invented employees; resets on page reload
Database (Supabase)	**Designed, not applied**	4 migration files ready; the Supabase project has not been created
Sign-in	**Not built yet**	A role switcher (Employee / Manager / HR) stands in until Google sign-in is configured
Automated code checks	**Pass**	Type check, lint and production build all pass on a fresh copy from GitHub

2. What the app does

Section	What employees, managers and HR can do
Home	Today's shift and hours, one big Clock In / Clock Out button, Punch, who is off this week, birthdays and anniversaries, announcements, holidays
Inbox	One queue for everything waiting: leave, requests, documents to sign — approve, reject or comment in place
Attendance & Leave	Month calendar with day detail, regularisation requests, leave balances with the full working shown, apply for leave, team calendar
Me	Time, finances (marked "not configured"), documents, assets, ID card, my working rules, **face for attendance**
My Team / Directory	Departments, who is away, profiles, reporting tree, employment history and timeline
Policies & Wall	Policies with read-acknowledgement; company feed with reactions and comments
Employee Voice	Confidential grievances: anonymous tickets store no identity, harassment cases go only to the committee, managers have no access by design

Ask HR assistant	Answers from published policies and the person's own records, cites the policy, refuses on grievances, pay disputes and legal questions
Reports & Surveys	Headcount, attendance, leave and tenure reports; a report builder; eNPS surveys (no result shown below 5 respondents)
Settings (HR)	Custom fields, approval chains, integrations, **attendance sites** (new)

3. New: location + face check on attendance

Clock In, Clock Out and Punch now save **only after both checks**:

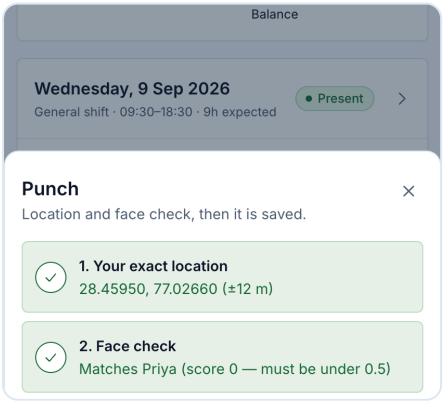
1. **Exact location** — the phone's GPS (high accuracy, never a cached reading). No location, no punch.
2. **Face check** — a selfie must match the face the employee registered once (the first time they clock in, or from Me → Face for attendance, with a consent checkbox). A different face is **blocked**.

If the person is farther from their site than allowed (default 200 m), the punch is **saved but flagged** for the manager, never blocked, because installers clock in at client sites. HR records each site's position by standing there and tapping "use my location" (Settings → Attendance sites). Each punch keeps the location, its accuracy, the distance from the site, the match score and a small selfie for review.

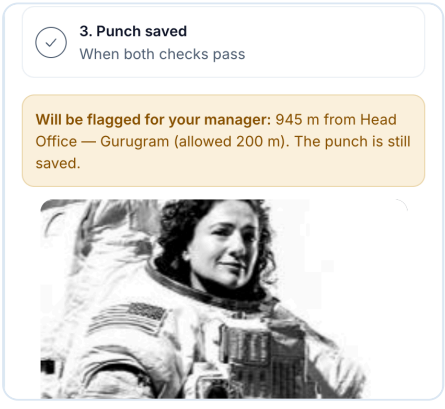
Face matching runs **entirely on the phone** (open-source face-api library); no photo is sent to any server in this version.

Test results (21 Oct 2026, real Chrome with a simulated camera and GPS)

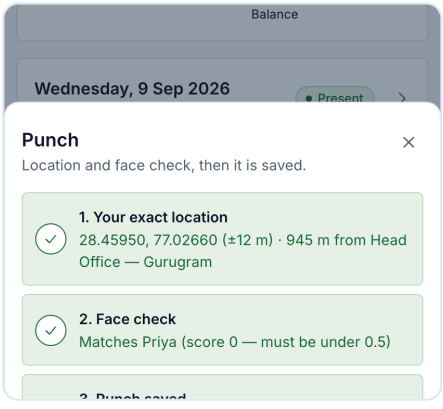
Test	Expected	Result
Register a face, then punch as the same person	Saved with location and match	**Passed**
A different person tries to punch for that employee	Blocked, nothing saved	**Passed**
Punch 945 m from the site (limit 200 m)	Saved and flagged for the manager	**Passed**
Location permission refused	Clear message, nothing saved	**Passed**
Camera permission refused	Clear message, nothing saved	**Passed**



Same person — saved



Different person — blocked



Far from site — saved, flagged

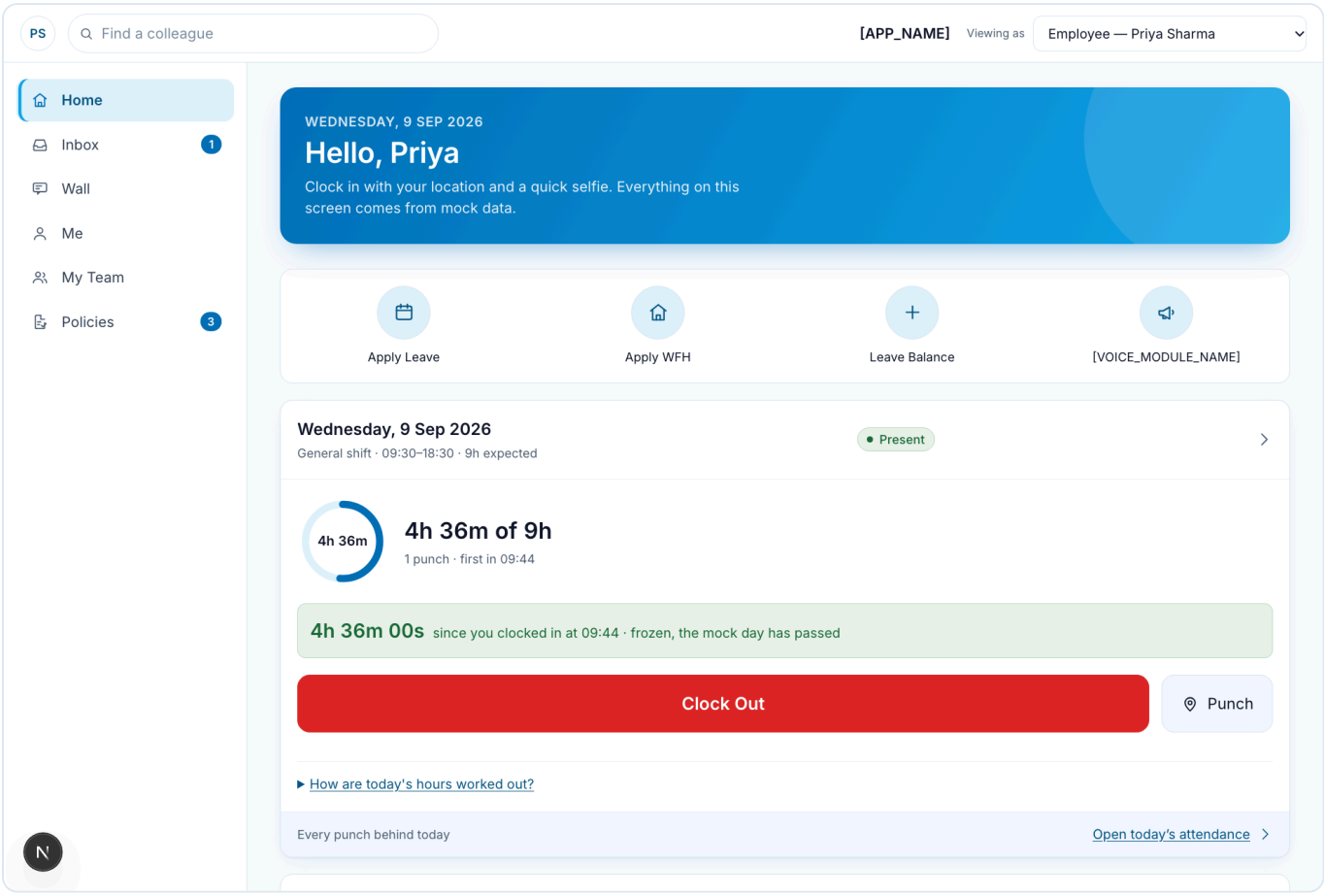
Before using this with real staff (details in docs/13, questions 28–33):

- **Biometric data (DPDP Act):** consent wording, how long face data is kept, who can delete it, and where it is stored need an HR/legal decision.
- **Liveness:** a printed photo or a phone screen held up to the camera could pass. Add a blink/turn-head check before go-live.
- **Strictness and distance:** the face-match limit (0.5) and the 200 m radius are starting values, not yet tuned on real phones and staff.

- **Staff without a camera or GPS** currently cannot clock in, so a fallback is needed.
- Who reviews flagged punches, and whether a manager can reject one.

4. New look

Changed from brown and beige to **white with sky blue**: white sidebar and header, a sky-blue active menu item, a sky-blue welcome banner on Home, rounded cards with soft shadows, and pill-shaped buttons. All text colours were checked for readability (WCAG AA contrast).



5. How to run it

```
git clone https://github.com/jiyasiwach/magppie-hr-app.git
cd magppie-hr-app
git checkout verified-punch-and-sky-theme   # this round's work (local branch until pushed)
npm install
npm run dev -- -p 3040                      # then open http://localhost:3040
```

Use the "Viewing as" switcher (top right) to see the Employee, Manager and HR views. To try the camera feature, open the app in **Chrome or Safari** and allow location and camera when asked. On a phone it needs HTTPS (the deployed site), because browsers only allow camera and GPS on secure pages.

6. What is deliberately not real yet

- All 40 employees and their records are invented, and changes reset when the page reloads.
- No file storage: documents are records, and "Download" explains there is nothing behind it.
- Salary and Expenses show "not configured". There is no payroll.
- The feedback form says plainly that nothing was sent.
- The product name is not chosen. The app shows `[APP_NAME]` (a one-line change).

7. Decisions and blockers

#	Needed	Who / why
1	Create the Supabase project (`magppie-hr-prod`)	Owner. Nothing on the database side can start without it.
2	Choose the product name	Management
3	Move the GitHub repo from `jiyasiwach` to `Magppie1234`	Owner. This unlocks automatic deploys on the Magppie Vercel team.
4	Configure Google sign-in	Google Workspace admin. Also confirm factory and site staff have Workspace accounts.
5	HR policy answers (33 open questions)	HR: shift timings, who is HR admin, leave rules, approval chains, document visibility, and the face/location questions above
6	Data-protection basis	HR/legal: the database region (Singapore) holds Indian employees' data, plus the new biometric consent

8. Recommended next steps

1. Create Supabase and apply the 4 migrations; add tables for face registrations and attendance sites.
2. Connect the screens to the database one module at a time (the code is structured for this swap).
3. Add Google sign-in and test access rules with real Employee, Manager and HR sessions.
4. Add a liveness check to the face step; get the DPDP consent wording approved.
5. Add a manager review queue for flagged punches.
6. Pilot with one office and one factory before replacing Keka.

9. Where things are

Item	Location
Code (temporary home)	github.com/jiyasiwach/magppie-hr-app — this round is on branch `verified-punch-and-sky-theme` , not yet pushed
Live site	magppie-hr-app.vercel.app (personal Vercel account, temporary). Does **not** include this round's changes yet.
Full technical docs	`AI_MEMORY.md` (start here), `README.md` , and `docs/01–17` : architecture, database, business rules, workflows, testing, open questions, backlog
Database design	`supabase/migrations/0001–0004`
New feature code	`src/components/attendance/` , `src/data/verification.ts` , `src/lib/face.ts` , `src/lib/geo.ts` , `/me/face` , `/settings/sites`